

Supervised Covariance Reset Resolves the Tracking–Confidence Tradeoff in Topology-Synchronous Capacitor Health Estimation

Anonymous Author(s)

Abstract—Topology-synchronous decoupled observations identify the capacitance and equivalent series resistance (ESR) of the Ćuk energy-transfer capacitor from normal switching, but their two published realizations trade point tracking against confidence calibration: a direction-decoupled recursive least-squares (RLS) estimator tracks and recovers quickly with uncalibrated intervals, whereas the Kalman realization reports calibrated intervals but freezes after abrupt health changes because its innovation gate then rejects every row. This letter adds a per-direction two-sided CUSUM supervisor that consumes both accepted and gate-rejected normalized innovations and, on alarm, resets that direction’s covariance to its initialization value. On identical simulated observation streams over a 54-case blind-style matrix at two noise levels, the supervised estimator matches its Kalman parent in steady state (joint 95% coverage 96.9%/93.4%; false-alarm cost at most 0.05 percentage points), recovers from abrupt -20% capacitance and $2\times$ ESR steps in 2–3 switching cycles instead of 527–868 (RLS) or never (Kalman), and reduces the hardest 0.1-s ramp capacitance tracking error below both parents, at about six extra operations per row. Results are simulation-only; the components are classical, and the contribution is the gated topology-synchronous construction that removes the documented tradeoff.

Index Terms—Capacitor condition monitoring, Ćuk converter, change detection, Kalman filters, recursive least squares.

I. INTRODUCTION

Capacitance loss and ESR growth are established degradation precursors in power converters [1], [2], and joint C–ESR estimation from converter waveforms is a mature line of work [3], [4]. For the Ćuk energy-transfer capacitor, the companion paper [5] constructs direction-decoupled, topology-synchronous observations—an ESR-dominant switching-edge row and a capacitance-dominant safe-window charge row—and evaluates two realizations on a common gated observation stream: TS-D-RLS, a direction-specific scalar RLS pair, and TS-SLTVKE, a structured Kalman estimator with normalized-innovation-squared (NIS) gating.

That evaluation left a documented tradeoff. TS-D-RLS delivered the better tracking, convergence, and abrupt-step recovery, but its auxiliary intervals covered only 13% jointly; TS-SLTVKE delivered calibrated intervals (88.9%) but failed to recover from abrupt health steps: once the parameter jumps, every innovation exceeds the NIS gate, the update is rejected, the covariance stays small, and the estimate freezes at the stale value. The companion paper retained this failure as a negative result and named an adaptive change detector as future work.

This letter closes that loop. The proposed construction leaves the Kalman realization untouched in steady state and

adds a supervisor built from classical components—Page’s CUSUM statistic [6] and covariance resetting from adaptive filtering practice [7], [8], [9], [10]. The mechanism observation is that the gate-rejected innovations, which the point update discards, carry exactly the sustained same-sign evidence of a parameter change; supervising them recovers RLS-class adaptation without giving up the Kalman covariance readout. Neither component is claimed as new; the contribution is their direction-specific use on gated topology-synchronous rows and the demonstrated resolution of the tracking–confidence tradeoff.

II. SUPERVISED-RESET CONSTRUCTION

Each accepted observation row is scalar in one health direction $j \in \{C, R\}$ [5]: $z_{j,k} = h_{j,k}\theta_j + \nu_{j,k}$, with $\theta_C = \bar{\alpha}$ (normalized inverse capacitance) and $\theta_R = r_C$. The Kalman kernel per direction is the standard scalar predict–gate–update recursion with process variance Q_j per switching cycle, measurement variance R_j , innovation $\tilde{z} = z_{j,k} - h_{j,k}\hat{\theta}_j$, and normalized innovation

$$\varepsilon_{j,k} = \tilde{z} / \sqrt{S_{j,k}}, \quad S_{j,k} = h_{j,k}^2 P_j + R_j, \quad (1)$$

accepted when $\varepsilon_{j,k}^2 \leq \gamma$ with the predeclared gate $\gamma = 9$.

The supervisor maintains a two-sided CUSUM per direction over *every* valid row, accepted or rejected, using the clipped statistic $\bar{\varepsilon} = \text{clip}(\varepsilon_{j,k}, \pm c)$:

$$g_j^+ \leftarrow \max(0, g_j^+ + \bar{\varepsilon} - \kappa), \quad g_j^- \leftarrow \max(0, g_j^- - \bar{\varepsilon} - \kappa). \quad (2)$$

When $\max(g_j^+, g_j^-) > h_a$, the supervisor resets $P_j \leftarrow P_{j,0}$ (the initialization variance) and zeros both sums. All constants are predeclared and shared by both directions: $\kappa = 0.5$, $h_a = 10$, $c = 6$. Under no change, $\varepsilon_{j,k}$ is approximately zero-mean with unit variance, so the drift κ keeps both sums near zero at any calibrated noise level; after an abrupt change, the rejected innovations saturate at $\pm c$ and the alarm fires within a few valid rows. The reset restores the Kalman gain, the estimate re-converges as from initialization, and the reported interval widens and then re-tightens—an honest reflection of the post-change uncertainty. The supervisor costs two comparisons and about six additions/multiplications per valid row on top of the Kalman kernel (46 operations in the companion accounting).

III. EVALUATION PROTOCOL

All estimators consume identical O1 feature streams generated from the companion switched Ćuk model (transfer-capacitor ESR retained, other parasitics zero; 24 V, $D = 0.40$,

TABLE I
STATIC BLIND-STYLE MATRIX (54 RUNS PER CELL)

Noise	Method	C MAPE (%)	ESR MAPE (%)	Conv. (cycles)	Joint cov. (%)
nominal	TS-D-RLS	0.160	0.050	5.4	—
nominal	Kalman kernel	0.118	0.143	3.5	96.9
nominal	supervised	0.120	0.143	3.4	96.9
10 mV/5 mA	TS-D-RLS	1.611	0.289	92.5	—
10 mV/5 mA	Kalman kernel	1.069	0.473	118.7	93.4
10 mV/5 mA	supervised	1.121	0.473	119.4	93.4

The TS-D-RLS auxiliary interval is uncalibrated [5] and its coverage is omitted.

50 kHz, $C_b = 100 \mu\text{F}$, $r_b = 50 \text{ m}\Omega$) with the locked gates, projection bounds, and hyperparameters of [5]. The static matrix crosses $C/C_b \in \{0.8, 0.9, 1.0\}$, $r_C/r_b \in \{1, 1.5, 2\}$, load factors $\{0.58, 1, 1.45\}$, and two voltage/current noise profiles (1 mV/0.5 mA and 10 mV/5 mA), with three seeds per cell (162 runs of 1500 cycles) and initialization drawn from the companion blind ranges. The edge gain k_R and empirical row variances are calibrated per noise profile on nominal health with disjoint seeds. Dynamic scenarios at nominal noise use three seeds each: a -20% capacitance step, a $2\times$ ESR step, and the companion's hardest 0.1-s joint ramp. Boundaries: simulation with ideal 10-MS/s sampling of the plant (not the F28379D chain); the compared Kalman column is the scalar two-direction kernel realizing the TS-SLTVKE point behavior; and noise-only intervals are not expected to cover during sustained drift.

IV. RESULTS

A. Steady State

Table I confirms the design goal: with only 11 supervisor resets across the 243 000 supervised static cycles (a false-alarm rate of 4.5×10^{-5} per cycle), the supervised estimator matches its Kalman parent to within 0.003 (nominal) and 0.05 (high-noise) percentage points of capacitance MAPE, with identical ESR error and joint coverage. TS-D-RLS keeps its known static ESR advantage; the latter claim is not uniform dominance but the removal of the tradeoff that forced a dual realization.

B. Abrupt Recovery and Ramp Tracking

Table II and Fig. 1 show the resolution of the tradeoff. After the capacitance and ESR steps the Kalman kernel never re-enters the 5% band within the 1200-cycle horizon (stale-value errors of 21.0% and 50.0%), TS-D-RLS recovers in 527 and 868 cycles, and the supervised estimator recovers in 3.3 and 2 cycles with one reset each and post-recovery joint coverage of 99.4%/98.5%. On the 0.1-s joint ramp the supervisor fires about six times, acting as an adaptive gain schedule: the capacitance tracking error (10.3% normalized RMSE) falls below both parents (19.2%/34.2%) while the ESR error retains the Kalman advantage (0.40% versus 12.8%). Interval coverage during the ramp itself degrades to about 30% because the noise-only covariance does not model drift bias; this boundary is inherent to both Kalman columns and is reported, not hidden.

TABLE II
DYNAMIC SCENARIOS AT NOMINAL NOISE (MEAN OF THREE SEEDS)

Scenario	Method	Recovery (cycles)	Tail err. (%)	Post cov. (%)
$C \rightarrow 0.8C$	TS-D-RLS	527	0.98	—
	Kalman kernel	>1200	20.99	99.7
	supervised	3.3	0.03	99.4
$r_C \rightarrow 2r_C$	TS-D-RLS	868	2.42	—
	Kalman kernel	>1200	50.02	99.4
	supervised	2	0.12	98.5
0.1-s ramp	TS-D-RLS	—	19.2/12.8 [†]	—
	Kalman kernel	—	34.2/0.40 [†]	—
	supervised	—	10.3/0.40[†]	—

[†]Normalized RMSE (%) for C/ESR over the ramp; recovery and post-step coverage do not apply. Kalman-column post-step coverage is dominated by the pre-step settled window.

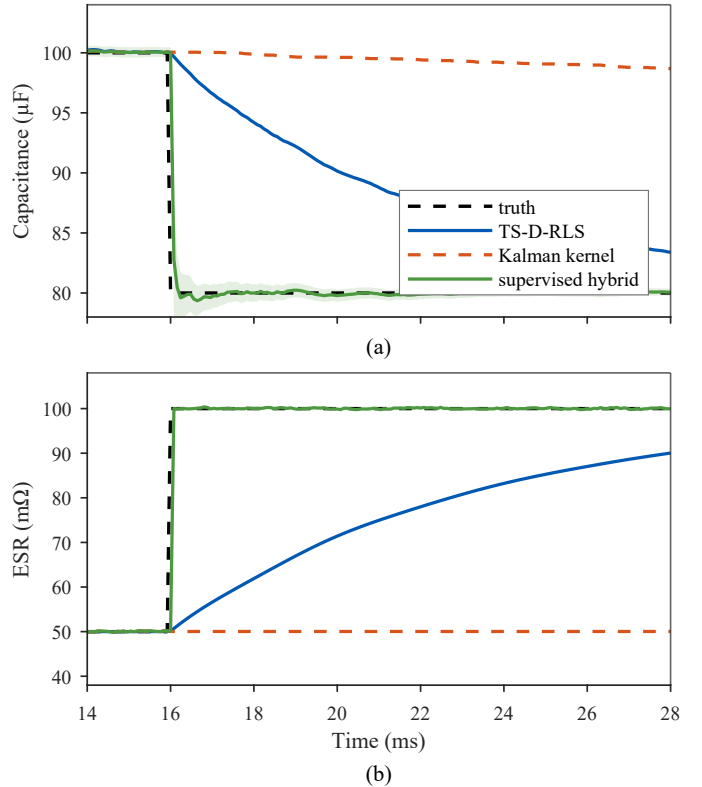


Fig. 1. Abrupt-step recovery on identical observation streams (seed 1): (a) capacitance estimates after the -20% step; (b) ESR estimates after the $2\times$ step. The shaded band is the supervised estimator's 95% interval, which widens at the reset and re-tightens. The Kalman kernel freezes at the stale value; TS-D-RLS recovers over hundreds of cycles.

V. CONCLUSION

A predeclared two-sided CUSUM supervisor over accepted and gate-rejected normalized innovations, with a per-direction covariance reset, lets a single topology-synchronous Kalman estimator keep calibrated steady-state intervals while recovering from abrupt capacitor-health changes in a few switching cycles and improving worst-case ramp tracking beyond both published realizations, at about six extra operations per row. The evidence is simulation-only and inherits the companion paper's plant and observation boundaries; hardware validation, DCM operation, and drift-aware interval inflation remain open.

REFERENCES

- [1] H. Wang and F. Blaabjerg, "Reliability of capacitors for DC-link applications in power electronic converters—an overview," *IEEE Transactions on Industry Applications*, vol. 50, no. 5, pp. 3569–3578, 2014.
- [2] Z. Zhao, P. Davari, W. Lu, H. Wang, and F. Blaabjerg, "An overview of condition monitoring techniques for capacitors in DC-link applications," *IEEE Transactions on Power Electronics*, vol. 36, no. 4, pp. 3692–3716, 2021.
- [3] K. Yao, W. Tang, W. Hu, and J. Lyu, "A current-sensorless online ESR and C identification method for output capacitor of buck converter," *IEEE Transactions on Power Electronics*, vol. 30, no. 12, pp. 6993–7005, 2015.
- [4] K. Hasegawa, S. Nishizawa, and I. Omura, "ESR and capacitance monitoring of a DC-link capacitor used in a three-phase PWM inverter with a front-end diode rectifier," *Microelectronics Reliability*, vol. 88–90, pp. 433–437, 2018.
- [5] Anonymous Author(s), "Topology-synchronous decoupled C–ESR identification for the energy-transfer capacitor in Ćuk converters," *submitted for publication*, 2026.
- [6] E. S. Page, "Continuous inspection schemes," *Biometrika*, vol. 41, no. 1/2, pp. 100–115, 1954.
- [7] R. K. Mehra, "On the identification of variances and adaptive Kalman filtering," *IEEE Transactions on Automatic Control*, vol. 15, no. 2, pp. 175–184, 1970.
- [8] A. S. Willsky, "A survey of design methods for failure detection in dynamic systems," *Automatica*, vol. 12, no. 6, pp. 601–611, 1976.
- [9] F. Gustafsson, *Adaptive Filtering and Change Detection*. Chichester, U.K.: John Wiley & Sons, 2000.
- [10] Y. Bar-Shalom, X. R. Li, and T. Kirubarajan, *Estimation with Applications to Tracking and Navigation: Theory, Algorithms and Software*. New York, NY, USA: Wiley-Interscience, 2001.